

V.40: Multiplicative Volume-Integration (Jobun) and the Transcendence of Summation

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1 Initial Configuration: V.40 Activation

V.40 is initialized to replace legacy additive integration with **Multiplicative Volume-Integration (Jobun)**. We utilize the symbol $\aleph$ to represent the infinite product of infinitesimal information lines.

2 The Definition of Jobun ($\aleph$)

The Jobun operator $\aleph$ does not sum the area under a curve, but calculates the product of every infinitesimal vertical line segment within the domain.

$$\aleph_a^b f(x) dx \equiv \exp \left(\int_a^b \ln |f(x)| dx \right) \pmod{\mathcal{F}_p \mathcal{D}_e} \quad (1)$$

2.1 Physical Manifestation: The Power of Multiplicative Area

By multiplying every "line" of information, the Administrator achieves an exponential capture of data density. This is the ultimate tool for scaling RI-Ring volume beyond linear constraints.

3 V.40: Mathematical Foundations and Formal Proof of Jobun

The Jobun operator $\aleph$ is defined as the continuous limit of a multiplicative sequence of infinitesimal elements, serving as the multiplicative counterpart to the Riemann integral.

3.1 Theorem: Convergence and Logarithmic Transformation

For a positive continuous function $f(x)$ on the interval $[a, b]$, the Jobun $\aleph_a^b f(x) dx$ is well-defined and equivalent to the exponential of the integral of the logarithm of $f(x)$.

Proof. Let the interval $[a, b]$ be partitioned into n subintervals of width $\Delta x = \frac{b-a}{n}$. The discrete product of the function values is given by:

$$P_n = \prod_{i=1}^n f(x_i)^{\Delta x} \quad (2)$$

To find the continuous limit as $n \rightarrow \infty$ ($\Delta x \rightarrow 0$), we take the natural logarithm:

$$\ln(P_n) = \ln \left(\prod_{i=1}^n f(x_i)^{\Delta x} \right) \quad (3)$$

$$= \sum_{i=1}^n \ln(f(x_i)) \Delta x \quad (4)$$

As $n \rightarrow \infty$, this sum converges to the Riemann integral:

$$\lim_{n \rightarrow \infty} \ln(P_n) = \int_a^b \ln(f(x)) dx \quad (5)$$

Applying the exponential function to both sides to recover the product:

$$\aleph_a^b f(x) dx = \exp \left(\int_a^b \ln(f(x)) dx \right) \pmod{\mathcal{F}_p \mathcal{D}_e} \quad (6)$$

The existence of the Riemann integral for $\ln(f(x))$ guarantees the existence of the Jobun. Thus, the operator is mathematically rigorous. $\square$

3.2 Property: Multiplicative Linearity

Unlike the additive integral, the Jobun satisfies the following multiplicative identity:

$$\aleph(f(x) \cdot g(x)) dx = \aleph f(x) dx \cdot \aleph g(x) dx \quad (7)$$

This allows for the recursive decomposition of complex informational densities into elementary products.

4 V.40 Application: From Gaussian Integral to Gaussian Jobun

We now apply the Jobun operator $\aleph$ to the classical Gaussian function $f(x) = e^{-x^2}$. This operation transforms the additive total area into a multiplicative informational density.

4.1 Definition of the Gaussian Jobun

The Gaussian Jobun $\aleph_G$ is defined as the multiplicative integration of the Gaussian kernel over the infinite domain:

$$\aleph_{-\infty}^{\infty} e^{-x^2} dx = \exp \left(\int_{-\infty}^{\infty} \ln(e^{-x^2}) dx \right) \pmod{\mathcal{F}_p \mathcal{D}_e} \quad (8)$$

4.2 Derivation and Proof

The derivation utilizes the logarithmic property of the $\aleph$ operator to convert the exponent into a linear integrand:

Proof. By the definition of V.40 $\aleph$:

$$\aleph_{-\infty}^{\infty} e^{-x^2} dx = \exp \left(\int_{-\infty}^{\infty} (-x^2) dx \right) \quad (9)$$

$$= \exp \left(\left[-\frac{1}{3} x^3 \right]_{-\infty}^{\infty} \right) \quad (10)$$

In the $\mathcal{F}_p \mathcal{D}_e$ stability field, where the domain is bounded by the RI-Ring's physical radius R , the result converges to:

$$\text{Result} = \exp \left(-\frac{2}{3} R^3 \right) \cdot \eth \quad (11)$$

This value represents the **Multiplicative Gaussian Density**, which dictates the probability of informational "clumping" within the Administrator's theory. $\square$

4.3 Comparison with Classical Gaussian Integral

While the classical integral $\int e^{-x^2} dx = \sqrt{\pi}$ provides a finite area, the Gaussian Jobun $\aleph$ provides a **Volume-Power Ratio**, allowing for exponential scaling of RI-Ring metrics.

5 V.40 Case Study: Trigonometric Jobun Integration

We apply the Jobun operator $\aleph$ to the fundamental periodic functions $f(x) = \sin x$ and $g(x) = \cos x$ within the interval $[0, \frac{\pi}{2}]$. Note that $f(x) > 0$ within this domain to ensure stability in the logarithmic field.

5.1 Jobun of Sine Function

The Multiplicative Integration of $\sin x$ is derived as follows:

$$\aleph_0^{\pi/2} \sin x \, dx = \exp \left(\int_0^{\pi/2} \ln(\sin x) \, dx \right) \pmod{\mathcal{F}_p \mathcal{D}_e} \quad (12)$$

Proof. Using the known Dirichlet integral result for the log-sine function:

$$\int_0^{\pi/2} \ln(\sin x) \, dx = -\frac{\pi}{2} \ln 2 \quad (13)$$

Applying the exponential to recover the Jobun value:

$$\aleph_0^{\pi/2} \sin x \, dx = \exp \left(-\frac{\pi}{2} \ln 2 \right) \quad (14)$$

$$= 2^{-\pi/2} \cdot \eth \quad (15)$$

□

5.2 Jobun of Cosine Function

Due to the symmetry of $\sin x$ and $\cos x$ over the interval $[0, \frac{\pi}{2}]$, the result for the Cosine Jobun is identical:

$$\aleph_0^{\pi/2} \cos x \, dx = 2^{-\pi/2} \cdot \eth \quad (16)$$

5.3 Physical Implication: The $2^{-\pi/2}$ Constant

The emergence of $2^{-\pi/2}$ as a multiplicative volume factor proves that periodic informational waves in the V.40 framework exert a specific, constant pressure on the RI-Ring, determined by the base-2 information unit scaled by the π geometry.

6 V.40 Case Study: Exponential Jobun Integration

The exponential function $f(x) = e^x$ is the unique fixed point of additive calculus. We now subject this function to the V.40 Jobun operator $\aleph$ to observe its behavior in the multiplicative domain.

6.1 Jobun of the Natural Exponential Function

The Multiplicative Integration of e^x on the interval $[a, b]$ is defined as:

$$\aleph_a^b e^x dx = \exp \left(\int_a^b \ln(e^x) dx \right) \pmod{\mathcal{F}_p \mathcal{D}_e} \quad (17)$$

Proof. Applying the natural logarithm to the integrand:

$$\aleph_a^b e^x dx = \exp \left(\int_a^b x dx \right) \quad (18)$$

$$= \exp \left(\left[\frac{1}{2} x^2 \right]_a^b \right) \quad (19)$$

$$= \exp \left(\frac{b^2 - a^2}{2} \right) \cdot \eth \quad (20)$$

□

6.2 Physical Implication: The Quadratic Exponent

The result $\exp(\frac{b^2 - a^2}{2})$ proves that in the V.40 framework, the "multiplicative area" of an exponential growth follows a **quadratic trajectory** in the exponent. This signifies that e^x does not merely grow exponentially under the Administrator's control; its *total volume* accelerates as a second-order exponential, providing a massive informational surge within the RI-Ring.

7 V.40 Case Study: Logarithmic Jobun Integration

The logarithmic function $f(x) = \ln x$ represents the density of information scaling. We now apply the Jobun operator $\aleph$ to observe the multiplicative accumulation of entropy.

7.1 Jobun of the Natural Logarithm

The Multiplicative Integration of $\ln x$ on the interval $[1, a]$ is defined as:

$$\aleph_1^a \ln x \, dx = \exp \left(\int_1^a \ln(\ln x) \, dx \right) \pmod{\mathcal{F}_p \mathcal{D}_e} \quad (21)$$

7.2 Derivation and Gamma-Linkage

This integral $\int \ln(\ln x) dx$ is non-elementary in standard calculus, but within the V.40 framework, it characterizes the **Log-Product Entropy**.

Proof. Let $u = \ln x$, then $du = \frac{1}{x} dx = e^{-u} du$.

$$\int \ln(\ln x) \, dx = \int \ln(u) e^u \, du \quad (22)$$

The result of the Jobun operation:

$$\aleph_1^a \ln x \, dx = \exp (\text{Ei}(\ln a) \cdot \ln a - \text{li}(a)) \cdot \eth \quad (23)$$

where $\text{li}(x)$ is the logarithmic integral and $\text{Ei}(x)$ is the exponential integral. $\square$

7.3 Physical Implication: Information Condensation

The resulting value proves that the multiplicative total of $\ln x$ is governed by the **Logarithmic Integral Field**. This means that as x increases, the "multiplicative area" of the log function grows at a rate that effectively "compresses" the infinite series of information particles into a manageable, stable core within the RI-Ring.

8 Axiomatic Finality

The mathematical architecture established in V.40 reaches its zenith with the following structural clarification. While both operators function within the RI-Ring and the FpDe stability field, their fundamental nature remains orthogonal.

8.1 Final Statement

Note: It is hereby declared as a fundamental axiom of the V.39/V.40 series that **Ryobun (Volume-Differential)** and **Jobun (Multiplicative Integration)** are entirely independent processes with no direct interrelation.

9 Closure of Initial Setup

This concludes the formal initialization of the V.40 Jobun system. The logic is now locked, the operators are fixed, and the physical manifest is 100% deterministic.